

Online Appendix

Artificial Intelligence, Algorithmic Recommendations and Competition

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This Online Appendix provides additional results and analytical details omitted from the main text. Sections are labeled A through I; references such as “Section 9” refer to sections of the main text.

A. RECOMMENDER SYSTEMS

This section provides additional details on the estimation and regularization procedures.

A.1. Estimation procedure

The minimization of the loss function (5) (or (6)) is performed numerically. We first calculate the gradient vector of partial derivatives with respect to the $J \times H + I \times H$ decision variables. In vector notation, the gradient is

$$\mathbf{G} = - \begin{pmatrix} \mathbf{E}\hat{\mathbf{V}} - \lambda_t \hat{\mathbf{T}} \\ \mathbf{E}'\hat{\mathbf{T}} - \lambda_v \hat{\mathbf{V}} \end{pmatrix}, \quad (\text{A.1})$$

where $\mathbf{E}$ is the $I \times J$ matrix of residuals $e_{ij} = \tilde{r}_{ij} - \sum_{h=1}^H \hat{t}_{ih} \hat{v}_{jh}$ on the observed entries. Starting from arbitrary initial values $\hat{\mathbf{T}}_0, \hat{\mathbf{V}}_0$, we use the Stochastic Gradient Descent to compute $\mathbf{G}_1, \dots, \mathbf{G}_t$ according to

$$\begin{pmatrix} \mathbf{T}_{t+1} \\ \mathbf{V}_{t+1} \end{pmatrix} = \begin{pmatrix} \mathbf{T}_t \\ \mathbf{V}_t \end{pmatrix} - \zeta \mathbf{G}_t, \quad (\text{A.2})$$

where $\zeta > 0$ is the step size.

For the numerical calculations, we use the open-source library of [Chin et al. \(2016\)](#). The iterative procedure is stopped after 50 iterations to avoid overfitting and reduce the computational burden.

In the content-based specification, where product attributes are observed and only $\hat{\mathbf{T}}$ is estimated, the gradient reduces to its first block.

A.2. Regularization and cross-validation

The regularization parameters λ_t and λ_v , as well as the step size ζ , are determined by 5-fold cross-validation, as is customary in machine learning. We partition the observed ratings into five equally sized subsets, use four of them to estimate the model, and compute the out-of-sample root mean squared prediction error (RMSE) on the fifth. The procedure is replicated changing the left-out set, and the average RMSE is computed. The entire procedure is then replicated over the grid $\lambda_t, \lambda_v, \zeta \in \{0.01, 0.05, 0.10, 0.15, 0.20\}$, and the values that minimize the average RMSE are selected.

A.3. Estimating the number of latent factors

In the results reported in the main text and this appendix, we assume the algorithm knows the true number of latent factors, H . As a robustness check, we allow the algorithm to choose the number of latent factors through a cross-validation procedure similar to that described above. Keeping the true number of factors fixed at its baseline value $H = 3$, we let the algorithm select $\hat{H}$ from the grid $\{1, 2, 3, 4, 5\}$. This procedure always selects $\hat{H} = 3$. The economic results are therefore unchanged.

B. ENDOGENOUS DATA

In the case of endogenous data, the matrix $\tilde{\mathbf{R}}$ is filled in gradually, in a number of successive steps indexed by $\tau = 0, 1, 2, \dots$. Initially, $\tilde{\mathbf{R}}_0$ is the empty matrix; that is, a matrix with all entries missing. At step $\tau = 1$, each consumer tries a randomly drawn product, resulting in ratings $\tilde{u}_{ij}$ for the RS. The matrix $\tilde{\mathbf{R}}_1$ therefore has a density of $1/J$.

At each subsequent step, the RS uses $\tilde{\mathbf{R}}_\tau$ to produce an estimate $\hat{\mathbf{R}}_\tau$ of the true matrix according to the estimation procedure described in Section 3 of the paper and recommends to each consumer i the product with the highest rating:

$$j^*(i, \tau) = \arg \max_j \hat{u}_{ij\tau}. \quad (\text{B.1})$$

This product is then tried by the consumer, and the corresponding rating is reported to the platform. In other words, during the learning phase we abstract from the possibility that consumers may engage in search. Search is introduced only after learning has been completed. Thus, $\tilde{\mathbf{R}}_{\tau+1}$ is obtained by adding to $\tilde{\mathbf{R}}_\tau$ one and only one entry $\tilde{u}_{ij^*(i, \tau)}$ for each of the I rows. If $\tilde{u}_{ij^*(i, \tau)}$ is not missing in $\tilde{\mathbf{R}}_\tau$, its value is overwritten. All other elements are unchanged.

As this process unfolds, the density of $\tilde{\mathbf{R}}_\tau$ increases. The learning phase ends when the matrix achieves a pre-specified density d .³⁵ At this juncture, we analyze the effects of the recommendations ultimately generated by the algorithm.

C. DEMAND FUNCTIONS

In this section, we provide additional details on the derivation of the demand functions and the calculation of total search costs, which depend on the expected number of inspections.

C.1. Preliminaries

To simplify notation, here we fix a consumer and suppress the consumer index. Thus, product j has systematic match value $\bar{u}_j$, price p_j , and idiosyncratic shock ε_j . The realized surplus from product j is

$$s_j = \bar{u}_j - p_j + \varepsilon_j.$$

³⁵In practice, achieving the target density level may require assuming that the algorithm operates in *exploration mode* for a certain fraction of periods, recommending products randomly. Without this, the process may reach a steady state with a density lower than the target. In our simulations, the case of endogenous data actually corresponds to the lowest possible probability of being in exploration mode, typically around 10 percent for $d = 1.2$. By increasing the probability of exploration mode, we can generate data that lie between purely exogenous and endogenous.

Remember that the shocks are i.i.d. Gumbel with location $-\gamma\beta$ and scale β , where γ denotes the Euler–Mascheroni constant, so that $E[\varepsilon_j] = 0$. The Gumbel assumption is convenient because Gumbel random variables with a common scale parameter β satisfy three useful properties. For any arbitrary scalar μ_j , let $\xi_j = \mu_j + \varepsilon_j$ and define the associated weights $w_j = e^{\mu_j/\beta}$. First, $\max_j \xi_j$ is itself Gumbel, with the same scale and aggregate weight $\sum_j w_j$. Second,

$$\Pr\left(\arg \max_j \xi_j = k\right) = \frac{w_k}{\sum_j w_j},$$

which is the familiar logit formula. Third, the maximum and the argmax are independent. These properties greatly simplify the derivation of demand.

C.2. Acceptability probabilities

In this subsection, we provide further details on the computation of the reservation surplus $\hat{s}$ and the acceptability probabilities.

Because consumers are symmetric ex ante, they all use the same cutoff. The reservation surplus $\hat{s}$ solves

$$\int \int \int [\bar{u} + \varepsilon - p - \hat{s}]_+ dF_{\bar{u}}(\bar{u}) dF_{\varepsilon}(\varepsilon) dF_p(p) = c_s, \quad (\text{C.1})$$

where $[x]_+ = \max\{x, 0\}$. Firms anticipate this cutoff when computing acceptability probabilities and, hence, their demand functions.

Under our assumptions, the left-hand side can be evaluated in three steps. First, under the degenerate price prior described in footnote 16 of the paper, the integral over prices collapses. Second, the expectation over the Gumbel shock has a closed-form expression. Let

$$a = \bar{u} - p - \hat{s} \quad \text{and} \quad z_0 = e^{a/\beta - \gamma}.$$

Using the identity

$$\int_0^x \frac{1 - e^{-z}}{z} dz = \gamma + \ln x - \text{Ei}(-x),$$

where $\text{Ei}(\cdot)$ denotes the exponential integral, we obtain

$$\begin{aligned} \int [a + \varepsilon]_+ dF_{\varepsilon}(\varepsilon) &= \int_{-a}^{\infty} [1 - \exp\{-e^{-(v+\gamma\beta)/\beta}\}] dv \\ &= \beta \int_0^{z_0} \frac{1 - e^{-z}}{z} dz \\ &= a - \beta \text{Ei}(-e^{a/\beta - \gamma}). \end{aligned} \quad (\text{C.2})$$

The original three-dimensional integral therefore reduces to a one-dimensional integral over $F_{\bar{u}}$. With $H = 3$ and conservative beliefs, $F_{\bar{u}}$ is uniform on $[-1, 1]$, and the integral can be calculated analytically. However, this is no longer possible under rational expectations, or with more general distributions of systematic match values. Therefore, we evaluate the remaining integral numerically before solving equation (C.1) for $\hat{s}$.

Finally, the acceptability probabilities follow directly from the Gumbel distribution function:

$$q_j = \Pr(s_j > \hat{s}) = 1 - \exp \left\{ -e^{(\bar{u}_j - p_j - \hat{s})/\beta - \gamma} \right\}. \quad (\text{C.3})$$

C.3. Demand under conservative beliefs

The derivation of demand formulas (11) and (12) in the main text relies on two convenient features of the model. First, with Gumbel shocks, the maximum surplus and the identity of the product attaining it are independent. Consequently, the argmax retains its logit distribution conditional on any event that depends only on the maximum, not merely on events of the form $\{\max_j s_j < \hat{s}\}$. Although the logit formula generally fails when conditioning on the acceptability of a particular product, it remains valid when conditioning on an event such as the existence of at least one acceptable product, since this event depends only on the maximum surplus.

Second, formula (12) is exact because rejection of the recommended product, $\{s_r \leq \hat{s}\}$, depends only on the shock ε_r and therefore conveys no information about the remaining products. Conditional on rejection, the consumer consequently searches randomly among the remaining $m - 1$ products, whose acceptability probabilities are unchanged. The same independence property makes the recursion for expected search length derived below exact. Neither of these simplifications generally carries over to the directed-search environment considered in Section 11.

C.4. Multiple recommendations

In this subsection, we provide the derivation of demand for the case of multiple recommendations, which is omitted in the main text. Let us assume, as in Section 8.5, that the consumer receives an ordered list $R = (r_1, \dots, r_C)$ of C products, ranked by estimated match value, inspects them in that order, and reverts to random search over the $m - C$ unlisted products if no listed product is acceptable. Listed product r_k is inspected if and only if all products ranked before it are rejected, so its fresh demand is

$$q_{r_k} \prod_{\ell < k} (1 - q_{r_\ell}), \quad k = 1, \dots, C. \quad (\text{C.4})$$

For an unlisted product $j \notin R$, fresh demand requires that every listed product be rejected, which occurs with probability $\prod_{\ell=1}^C (1 - q_{r_\ell})$, after which the argument of the previous subsection applies to the $m - C$ unlisted products:

$$\left[\prod_{\ell=1}^C (1 - q_{r_\ell}) \right] q_j \sum_{z=0}^{m-C-1} \frac{1}{1+z} B(z; \mathbf{q}_{-(R \cup \{j\})}). \quad (\text{C.5})$$

Returning demand is again unchanged.

C.5. Demand under rational expectations

The reliance effect arises when consumers update the distribution of systematic match values used to compute the search cutoff $\hat{s}$. Under rational expectations, this posterior coincides with the true distribution generated by the algorithm, which we obtain numerically.

For the intermediate cases between conservative beliefs and rational expectations discussed at the end of Section 10, we construct the cutoff as a convex combination of the two benchmark cutoffs, varying the weight placed on the rational-expectations cutoff.

C.6. Expected number of inspections

To compute consumer surplus net of search costs, we need the expected number of products inspected, denoted by $\mathbb{E}[T]$. Fortunately, the model admits exact representations, which may also be of independent interest.

Conditional on systematic match values and prices, product acceptability is independent across products. The derivations below rely only on this independence; the Gumbel assumption enters only through the acceptability probabilities q_j .

C.6.1. Random search

Let

$$A_j = \mathbf{1}\{s_j > \hat{s}\}$$

be the indicator that product j is acceptable, and let

$$K = \sum_{j=1}^m A_j$$

denote the total number of acceptable products. Conditional on $K \geq 1$, the expected position of the first acceptable product in a uniformly random permutation of the m products is

$$\mathbb{E}[T \mid K = k] = \frac{m+1}{k+1}.$$

If $K = 0$, the consumer inspects all products, so $\mathbb{E}[T \mid K = 0] = m$. Although products with larger q_j 's are more likely to be acceptable, conditional on the set of acceptable products the inspection order is uniform and independent of acceptability. Hence only the number K matters for the expected stopping time. Combining these observations gives

$$\mathbb{E}[T] = (m+1)\mathbb{E}\left[\frac{1}{K+1}\right] - \Pr(K=0), \quad (\text{C.6})$$

where

$$\mathbb{E}\left[\frac{1}{K+1}\right] = \int_0^1 \prod_{j=1}^m (1 - q_j + q_j x) dx,$$

and

$$\Pr(K=0) = \prod_{j=1}^m (1 - q_j).$$

The integral representation follows from

$$\frac{1}{k+1} = \int_0^1 x^k dx$$

and the probability-generating function of the Poisson-binomial distribution.

An alternative exact representation makes the connection with ordered search more transparent. Let

$$K_{-j} = \sum_{i \neq j} A_i$$

denote the number of acceptable products other than j . Product j is inspected if and only if it appears before every acceptable rival. Conditional on K_{-j} , this event has probability $1/(K_{-j} + 1)$. Therefore,

$$\mathbb{E}[T] = \sum_{j=1}^m \mathbb{E} \left[\frac{1}{K_{-j} + 1} \right] = \sum_{j=1}^m \int_0^1 \prod_{i \neq j} (1 - q_i + q_i x) dx. \quad (\text{C.7})$$

If no product is acceptable, all products are inspected, so the rule governing the consumer's final purchase does not affect T .

C.6.2. Single recommendation

Now consider a single recommendation. Let r denote the recommended product, which is inspected first. If it is acceptable, search stops after one inspection. If it is rejected, the consumer continues by searching randomly among the remaining $m - 1$ products. Let $\mathbb{E}_{-r}[T]$ denote the expectation in (C.6) applied to these non-recommended products. Then

$$\mathbb{E}[T] = 1 + (1 - q_r) \mathbb{E}_{-r}[T]. \quad (\text{C.8})$$

This recursion is exact because rejection of the recommended product, $\{s_r \leq \hat{s}\}$, depends only on ε_r and therefore leaves the acceptability probabilities of the remaining products unchanged. This conditional-independence property generally fails in the directed-search environment considered in Section 11.

C.6.3. Multiple recommendations

Finally, suppose that the platform provides an ordered list

$$R = (r_1, \dots, r_C).$$

Product r_k is inspected if all preceding recommended products have been rejected, an event with probability

$$\prod_{\ell < k} (1 - q_{r_\ell}).$$

The expected number of inspections within the recommendation list is therefore

$$\mathbb{E}[T_R] = \sum_{k=1}^C \prod_{\ell < k} (1 - q_{r_\ell}).$$

If all C recommended products are rejected, the consumer continues with random search among the $m - C$ unlisted products. Let $\mathbb{E}_{-R}[T]$ denote the expectation in (C.6) applied to this set. The total expected number of inspections is

$$\mathbb{E}[T] = \sum_{k=1}^C \prod_{\ell < k} (1 - q_{r_\ell}) + \prod_{\ell=1}^C (1 - q_{r_\ell}) \mathbb{E}_{-R}[T], \quad (\text{C.9})$$

where an empty product is understood to equal one. The first term in (C.9) can equivalently be written as

$$\sum_{k=1}^C \prod_{\ell < k} (1 - q_{r_\ell}) = \sum_{k=1}^C k q_{r_k} \prod_{\ell < k} (1 - q_{r_\ell}) + C \prod_{\ell=1}^C (1 - q_{r_\ell}).$$

The first term on the right-hand side accounts for stopping at the k -th recommended product, while the second accounts for rejecting the entire list.

D. NON-PERSONALIZED RECOMMENDATIONS

In this section, we consider the two types of non-personalized recommendations mentioned in footnote 23. First, we study an algorithm constrained to recommend the same product to every consumer. Specifically, it recommends the product with the highest average estimated match value (Table D.1). Second, we consider a popularity-based rule that recommends the product with the largest market share under random search (Table D.2).

m	Absolute gains			Percentage gains		
	7	13	19	7	13	19
Collaborative-filtering RS	0.0015 (0.0003)	0.0027 (0.0006)	0.0037 (0.0007)	0.34 (0.06)	0.54 (0.11)	0.73 (0.14)
Content-based RS	0.0016 (0.0003)	0.0027 (0.0006)	0.0036 (0.0007)	0.36 (0.06)	0.56 (0.11)	0.71 (0.14)

TABLE D.1: Changes in consumer surplus with algorithmic non-personalized recommendations.

Holding prices fixed, both types of non-personalized recommendations generate much smaller welfare gains than personalized recommendations. The uniform algorithmic recommendation is nearly valueless, while the popularity-based rule captures only a modest fraction of the gains from personalization. As discussed in the main text, this reflects the purely horizontal nature of product differentiation in our framework: no product is systematically preferred by all consumers, leaving little scope for non-personalized recommendations to improve matching.

m	Absolute gains			Percentage gains		
	7	13	19	7	13	19
Popularity-based recommendations	0.0040 (0.0002)	0.0092 (0.0003)	0.0133 (0.0004)	0.84 (0.05)	1.82 (0.07)	2.62 (0.09)

TABLE D.2: Changes in consumer surplus with popularity-based recommendations.

E. DECOMPOSITION OF GAINS

Table E.1 decomposes the total gains from algorithmic recommendations at fixed prices, reported in Section 7, into gains from improved matching and savings in search costs. Search-cost savings account for approximately 56% to 61% of the total gain, with their share declining as market size increases.

m	Collaborative RS			Content Based RS		
	7	13	19	7	13	19
Total gain	0.0810 (0.0008)	0.1657 (0.0013)	0.2236 (0.0011)	0.0822 (0.0008)	0.1688 (0.0013)	0.2285 (0.0011)
Improved matching	0.0319 (0.0003)	0.0692 (0.0005)	0.0975 (0.0006)	0.0327 (0.0003)	0.0713 (0.0005)	0.1008 (0.0006)
Search cost saving	0.0492 (0.0007)	0.0965 (0.0008)	0.1261 (0.0006)	0.0496 (0.0007)	0.0975 (0.0008)	0.1276 (0.0006)
% Improved matching/Total gain	39.46 (0.40)	41.81 (0.16)	43.59 (0.11)	39.89 (0.39)	42.29 (0.15)	44.13 (0.11)
% Search cost saving/Total gain	60.54 (0.40)	58.19 (0.16)	56.41 (0.11)	60.11 (0.39)	57.71 (0.15)	55.87 (0.11)

TABLE E.1: Surplus gain decomposition at fixed prices.

F. ROBUSTNESS ANALYSIS

In this section, we conduct comparative statics of the equilibrium under conservative beliefs with respect to parameters that do not directly affect the quantity or quality of information available to the algorithms. The corresponding analysis for information-related parameters is reported in Section G of this appendix.

F.1. Search cost

First, we vary the unit search cost c_s over a grid from 0.01 to 0.07, with the baseline value $c_s = 0.04$. Table F.1 maps each value of c_s into the equilibrium share of consumers who purchase the recommended product without further search. Tables F.2 and F.3 report the corresponding changes in equilibrium prices and consumer surplus relative to the random-search benchmark. Here and in the tables that follow, a subscript 99 (respectively, 98) indicates that only 99 (respectively, 98) of the 100 simulation runs converged to a pure-strategy price equilibrium.

As explained in the main text, the effect of recommendations on equilibrium prices becomes weaker as search costs fall and vanishes as c_s approaches zero. In this limit, consumers inspect all products before purchasing, so the order of inspection becomes irrelevant.

The effect of search costs on consumer surplus is non-monotone. When $c_s = 0$, recommendations have no effect on consumer surplus. At low levels of c_s , their effect is positive and increases with the search cost. Consumer surplus peaks at around $c_s = 0.03$ – 0.035 and declines thereafter. For $c_s \geq 0.05$, the increase in equilibrium prices more than offsets the gains from improved matching and lower search costs, so recommendations reduce consumer surplus relative to random search.

Search Cost	Collaborative RS			Content-based RS		
	7	13	19	7	13	19
0.01	0.200 (0.002)	0.259 (0.002)	0.296 (0.001)	0.202 (0.002)	0.262 (0.002)	0.301 (0.001)
0.015	0.279 (0.003)	0.363 (0.003)	0.415 (0.002)	0.281 (0.003)	0.368 (0.003)	0.422 (0.002)
0.02	0.347 (0.004)	0.454 (0.003)	0.516 (0.002)	0.350 (0.004)	0.460 (0.003)	0.525 (0.002)
0.025	0.407 (0.005)	0.533 (0.004)	0.601 (0.002)	0.411 (0.004)	0.540 (0.004)	0.612 (0.002)
0.03	0.461 (0.005)	0.601 (0.004)	0.672 (0.003)	0.465 (0.005)	0.610 (0.004)	0.685 (0.002)
0.035	0.509 (0.006)	0.659 (0.004)	0.730 (0.003)	0.513 (0.006)	0.669 (0.004)	0.746 (0.002)
0.04	0.552 (0.006)	0.709 (0.004)	0.778 (0.003)	0.556 (0.006)	0.720 (0.004)	0.795 (0.002)
0.045	0.591 (0.007)	0.751 (0.005)	0.817 (0.003)	0.596 (0.006)	0.762 (0.004)	0.835 (0.002)
0.05	0.625 (0.007)	0.786 (0.005)	0.849 (0.003)	0.631 (0.007)	0.798 (0.004)	0.867 (0.002)
0.055	0.657 (0.007)	0.816 (0.004)	0.874 (0.002)	0.662 (0.007)	0.828 (0.004)	0.893 (0.002)
0.06	0.685 (0.007)	0.841 (0.004)	0.894 (0.002)	0.690 (0.007)	0.853 (0.004)	0.913 (0.002)
0.065	0.710 (0.007)	0.862 (0.004)	0.911 (0.002)	0.715 (0.007)	0.874 (0.004)	0.930 (0.002)
0.07	0.732 ₉₉ (0.007)	0.879 (0.004)	0.924 (0.002)	0.738 (0.007)	0.892 (0.004)	0.943 ₉₈ (0.002)

TABLE F.1: Probability of purchasing the recommended product.

Search Cost	Collaborative RS			Content-based RS		
	7	13	19	7	13	19
0.01	0.45	0.74	1.07	0.49	0.78	1.16
	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)	(0.01)
0.015	1.27	2.14	3.08	1.38	2.31	3.41
	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)
0.02	2.54	4.46	6.40	2.75	4.86	7.21
	(0.08)	(0.07)	(0.08)	(0.07)	(0.06)	(0.06)
0.025	4.25	7.81	11.22	4.60	8.60	12.87
	(0.12)	(0.13)	(0.15)	(0.10)	(0.11)	(0.12)
0.03	6.43	12.32	17.78	6.94	13.70	20.74
	(0.18)	(0.21)	(0.26)	(0.15)	(0.18)	(0.21)
0.035	9.09	18.14	26.30	9.79	20.34	31.27
	(0.26)	(0.33)	(0.41)	(0.22)	(0.29)	(0.34)
0.04	12.23	25.38	37.04	13.14	28.70	44.95
	(0.35)	(0.50)	(0.64)	(0.31)	(0.45)	(0.55)
0.045	15.85	34.13	50.22	16.99	38.93	62.32
	(0.47)	(0.74)	(0.96)	(0.42)	(0.67)	(0.85)
0.05	19.93	44.49	66.04	21.32	51.18	84.00
	(0.61)	(1.04)	(1.40)	(0.56)	(0.97)	(1.28)
0.055	24.44	56.54	84.71	26.12	65.59	110.65
	(0.77)	(1.45)	(1.99)	(0.72)	(1.39)	(1.90)
0.06	29.37	70.40	106.36	31.35	82.33	143.09
	(0.96)	(1.97)	(2.76)	(0.92)	(1.93)	(2.74)
0.065	34.69	86.22	131.25	36.99	101.66	182.28
	(1.18)	(2.63)	(3.77)	(1.15)	(2.63)	(3.86)
0.07	40.35 ₉₉	104.11	159.67	43.04	123.87	228.49 ₉₈
	(1.44)	(3.47)	(5.06)	(1.40)	(3.50)	(5.34)

TABLE F.2: Percentage changes in equilibrium prices.

Search Cost	Collaborative RS			Content-based RS		
	7	13	19	7	13	19
0.01	0.0073 (0.0001)	0.0187 (0.0001)	0.0303 (0.0002)	0.0074 (0.0001)	0.0191 (0.0001)	0.0312 (0.0002)
0.015	0.0128 (0.0001)	0.0344 (0.0003)	0.0548 (0.0003)	0.0129 (0.0001)	0.0351 (0.0003)	0.0563 (0.0003)
0.02	0.0180 (0.0002)	0.0502 (0.0004)	0.0782 (0.0004)	0.0180 (0.0002)	0.0511 (0.0004)	0.0800 (0.0004)
0.025	0.0223 (0.0003)	0.0641 (0.0005)	0.0969 (0.0005)	0.0221 (0.0003)	0.0648 (0.0005)	0.0984 (0.0004)
0.03	0.0250 (0.0004)	0.0739 (0.0005)	0.1078 (0.0004)	0.0245 (0.0004)	0.0739 (0.0005)	0.1076 (0.0004)
0.035	0.0258 (0.0006)	0.0778 (0.0005)	0.1081 (0.0005)	0.0249 (0.0006)	0.0763 (0.0005)	0.1037 (0.0005)
0.04	0.0242 (0.0008)	0.0739 (0.0008)	0.0948 (0.0012)	0.0227 (0.0008)	0.0696 (0.0007)	0.0824 (0.0010)
0.045	0.0197 (0.0012)	0.0603 (0.0015)	0.0653 (0.0024)	0.0175 (0.0011)	0.0516 (0.0015)	0.0392 (0.0022)
0.05	0.0121 (0.0017)	0.0357 (0.0028)	0.0173 (0.0043)	0.0091 (0.0017)	0.0202 (0.0028)	-0.0308 (0.0042)
0.055	0.0012 (0.0024)	-0.0016 (0.0047)	-0.0518 (0.0071)	-0.0030 (0.0024)	-0.0265 (0.0048)	-0.1328 (0.0072)
0.06	-0.0133 (0.0034)	-0.0529 (0.0075)	-0.1438 (0.0111)	-0.0187 (0.0034)	-0.0904 (0.0077)	-0.2731 (0.0114)
0.065	-0.0314 (0.0046)	-0.1198 (0.0112)	-0.2613 (0.0164)	-0.0383 (0.0047)	-0.1740 (0.0116)	-0.4589 (0.0174)
0.07	-0.0532 ₉₉ (0.0061)	-0.2039 (0.0160)	-0.4070 (0.0236)	-0.0619 (0.0062)	-0.2798 (0.0166)	-0.6971 ₉₈ (0.0257)

TABLE F.3: Changes in consumer surplus at equilibrium prices, in model units.

F.2. Dispersion of the idiosyncratic shocks

Second, we vary the scale parameter β of the distribution of the idiosyncratic shocks ε_{ij} around its baseline value of 0.18. Tables F.4 and F.5 report the corresponding changes in equilibrium prices and consumer surplus.

The price effect of algorithmic recommendations weakens as the dispersion of the idiosyncratic shocks increases. When these shocks are more dispersed, consumers search more extensively, reducing the importance of the product at which search begins and hence the impact of personalized recommendations on firms' pricing incentives.

Accordingly, the gains in consumer surplus are smaller when the dispersion of the shocks is lower, because the resulting price increase is larger. For sufficiently low values of β , the price effect can more than offset the benefits from improved matching and reduced search costs, so that the RS lowers consumer surplus. At low values of β , however, convergence to a pure-strategy price equilibrium is not always obtained.

β	Collaborative RS			Content-based RS		
	7	13	19	7	13	19
0.135	27.04 (0.96)	74.40 (2.14)	128.13 ₉₈ (3.79)	27.80 (0.98)	85.22 (2.04)	161.66 ₉₉ (3.29)
0.158	18.29 (0.59)	43.38 (1.02)	65.22 (1.40)	19.33 (0.58)	50.31 (0.92)	84.69 (1.18)
0.18	12.06 (0.35)	25.99 (0.52)	37.95 (0.65)	12.86 (0.34)	29.68 (0.44)	46.29 (0.49)
0.203	7.78 (0.21)	15.76 (0.28)	22.12 (0.32)	8.42 (0.20)	18.10 (0.23)	27.02 (0.24)
0.225	5.14 (0.13)	10.17 (0.16)	14.15 (0.17)	5.55 (0.12)	11.50 (0.13)	16.78 (0.13)
0.248	3.45 (0.08)	6.79 (0.10)	9.40 (0.10)	3.71 (0.08)	7.59 (0.08)	10.95 (0.08)
0.27	2.33 (0.06)	4.62 (0.07)	6.38 (0.06)	2.53 (0.05)	5.17 (0.05)	7.43 (0.05)

TABLE F.4: Percentage changes in equilibrium prices.

β	Collaborative RS			Content-based RS		
	7	13	19	7	13	19
0.135	−0.0319 (0.0035)	−0.0868 (0.0070)	−0.2141 (0.0128)	−0.0342 (0.0036)	−0.1171 (0.0069)	−0.3157 (0.0121)
0.158	0.0032 (0.0018)	0.0216 (0.0025)	0.0069 (0.0039)	0.0007 (0.0019)	0.0065 (0.0024)	−0.0412 (0.0036)
0.18	0.0256 (0.0009)	0.0741 (0.0008)	0.0935 (0.0012)	0.0241 (0.0009)	0.0691 (0.0007)	0.0799 (0.0010)
0.203	0.0379 (0.0005)	0.0964 (0.0005)	0.1303 (0.0004)	0.0372 (0.0005)	0.0958 (0.0005)	0.1278 (0.0003)
0.225	0.0432 (0.0004)	0.1030 (0.0006)	0.1413 (0.0004)	0.0433 (0.0005)	0.1045 (0.0006)	0.1436 (0.0004)
0.248	0.0447 (0.0004)	0.1023 (0.0007)	0.1413 (0.0005)	0.0451 (0.0004)	0.1046 (0.0007)	0.1453 (0.0005)
0.27	0.0438 (0.0005)	0.0978 (0.0007)	0.1355 (0.0005)	0.0446 (0.0004)	0.1007 (0.0006)	0.1405 (0.0005)

TABLE F.5: Changes in consumer surplus at equilibrium prices, in model units.

F.3. Number of consumers and products

Third, we vary the number of consumers and products. We consider two departures from the baseline configuration. First, we double and halve both I and J , keeping the ratio $I/J = 30$ fixed. Second, we reduce the user-to-item ratio from 30 to 15 and then to 3, adjusting I and J so that the ratio ℓ of observations to estimated parameters remains constant. This second

exercise is motivated by the fact that benchmark datasets other than Netflix often feature lower user-to-item ratios. Tables F.6 and F.7 report the corresponding results.

Across all configurations, recommendations raise equilibrium prices and increase net consumer surplus relative to unassisted search, in line with the baseline findings.

I, J	Collaborative RS			Content-based RS		
	7	13	19	7	13	19
48000, 1600	10.18 (0.43)	20.52 (0.54)	26.69 (0.70)	13.45 (0.36)	31.58 (0.49)	49.68 (0.54)
12000, 400	8.02 (0.24)	15.73 (0.33)	20.84 (0.36)	9.93 (0.25)	21.46 (0.29)	31.82 (0.30)
12375, 825	11.69 (0.35)	24.92 (0.53)	35.40 (0.69)	12.95 (0.34)	29.94 (0.45)	46.83 (0.50)
3105, 1035	14.05 (0.37)	32.05 (0.57)	48.61 (0.76)	14.24 (0.38)	33.89 (0.52)	54.23 (0.61)

TABLE F.6: Percentage changes in equilibrium prices.

I, J	Collaborative RS			Content-based RS		
	7	13	19	7	13	19
48000, 1600	0.0286 (0.0010)	0.0799 (0.0008)	0.1077 (0.0010)	0.0235 (0.0010)	0.0676 (0.0008)	0.0756 (0.0012)
12000, 400	0.0275 (0.0006)	0.0726 (0.0005)	0.0973 (0.0005)	0.0261 (0.0007)	0.0713 (0.0004)	0.0907 (0.0005)
12375, 825	0.0264 (0.0009)	0.0756 (0.0008)	0.0974 (0.0012)	0.0240 (0.0009)	0.0690 (0.0007)	0.0792 (0.0010)
3105, 1035	0.0226 (0.0010)	0.0681 (0.0010)	0.0793 (0.0016)	0.0223 (0.0011)	0.0653 (0.0009)	0.0688 (0.0014)

TABLE F.7: Changes in consumer surplus at equilibrium prices, in model units.

F.4. Number of latent factors

Finally, we increase the number of latent factors to $H = 5$, which corresponds approximately to $H = 100$ in the Netflix calibration discussed in Section 6. The algorithm is still assumed to know the true number of factors. Tables F.8 and F.9 report the corresponding results.

The effects of recommendations with $H = 5$ closely mirror the baseline findings. Equilibrium prices increase by about 10% when $m = 7$ and by roughly 35% when $m = 19$. Consumer surplus also rises, with larger gains in larger markets.

Collaborative RS			Content-based RS		
7	13	19	7	13	19
10.06	22.60	34.59	10.72	24.72	38.71
(0.11)	(0.23)	(0.30)	(0.11)	(0.21)	(0.26)

TABLE F.8: Percentage changes in equilibrium prices with $H = 5$.

Collaborative RS			Content-based RS		
7	13	19	7	13	19
0.0331	0.0753	0.0951	0.0333	0.0756	0.0933
(0.0002)	(0.0002)	(0.0004)	(0.0002)	(0.0002)	(0.0004)

TABLE F.9: Changes in consumer surplus with $H = 5$.

G. INFORMATION AND WELFARE

In this section, we turn to comparative-statics exercises that vary the information available to the algorithm. Specifically, we vary the density d of the rating matrix, the amount of reporting noise, σ_ϵ^2 , the coarseness of the reporting scale, measured by the number k of Likert-scale levels, and the process generating the training data, from exogenous to endogenous data as described in Section 4.

As noted in the main text, outcomes depend more on the overall informativeness of the recommendations than on the particular source of variation. We measure this information level by consumer surplus when prices are fixed at marginal cost. Accordingly, Table G.1 reports, for each comparative-statics exercise and each parameter value, the resulting information level, which provides the horizontal coordinate in Figures 2 and 3 of the paper, together with the two outcomes of interest: equilibrium prices and consumer surplus. The table therefore provides both the detailed results for each exercise and the exact numerical values underlying the figures. We report results for $m = 13$; those for the other market sizes are qualitatively similar and are available in the replication package.

Exercise	Value	Info level	Collaborative RS		Content-based RS	
			$\Delta p\%$	ΔCS	$\Delta p\%$	ΔCS
d	0.6%	0.7900	14.74	0.0725	21.43	0.0713
d	0.8%	0.8036	21.32	0.0754	25.45	0.0712
d	1%	0.8088	24.20	0.0749	28.03	0.0702
d	1.2%	0.8117	25.99	0.0741	29.68	0.0691
d	1.4%	0.8137	27.40	0.0732	30.87	0.0682
d	1.6%	0.8153	28.43	0.0723	31.79	0.0674
d	1.8%	0.8167	29.79	0.0708	32.48	0.0668
d	2%	0.8177	30.77	0.0696	33.03	0.0662
d	2.2%	0.8185	31.45	0.0688	33.49	0.0657
d	2.4%	0.8191	31.92	0.0683	33.86	0.0653
σ_ε	0%	0.8137	27.25	0.0734	31.05	0.0680
σ_ϵ	10%	0.8131	26.80	0.0737	30.68	0.0683
σ_ϵ	20%	0.8117	25.99	0.0741	29.68	0.0691
σ_ϵ	30%	0.8088	24.01	0.0754	28.11	0.0703
σ_ϵ	40%	0.8057	22.30	0.0758	26.18	0.0714
σ_ϵ	50%	0.8016	20.00	0.0761	24.00	0.0722
σ_ϵ	60%	0.7976	17.93	0.0758	21.78	0.0725
σ_ϵ	70%	0.7932	15.76	0.0748	19.62	0.0722
σ_ϵ	80%	0.7889	13.84	0.0732	17.60	0.0713
σ_ϵ	90%	0.7850	12.21	0.0712	15.74	0.0699
k	2	0.7937	17.27	0.0733	19.08	0.0726
k	3	0.8043	22.52	0.0742	25.04	0.0719
k	5	0.8095	25.31	0.0737	27.97	0.0704
k	10	0.8116	26.49	0.0731	29.25	0.0695
Endog. data	—	0.8008	19.68	0.0749	26.59	0.0708

TABLE G.1: Comparative statics on information levels.

H. DIRECTED SEARCH

This section provides additional details on the directed-search environment analyzed in Section 11 of the paper. We derive demand, both without and with recommendations, and obtain closed-form expressions for the expected number of inspections.

H.1. Preliminaries

In the directed-search environment, consumers observe $\varepsilon_j - p_j$ for every product before inspection, but do not observe the systematic match values $\bar{u}_j$. Search is therefore governed by a reservation match value u^* , rather than by the reservation-surplus cutoff $\hat{s}$ used in the random-search environment. Define the ranking score

$$Z_j = u^* - p_j + \varepsilon_j$$

and the capped surplus

$$W_j = \min\{Z_j, s_j\} = \min\{\bar{u}_j, u^*\} - p_j + \varepsilon_j.$$

We also define

$$x_j = e^{-p_j/\beta}, \quad \lambda_j = e^{(\bar{u}_j - u^* - p_j)/\beta}, \quad c_j = e^{\min\{\bar{u}_j - u^*, 0\}/\beta},$$

and

$$S = \sum_{j=1}^m x_j c_j.$$

The derivations below repeatedly use two standard properties of independent Gumbel random variables with a common scale parameter. Let

$$\xi_j = \mu_j + \varepsilon_j \quad \text{and} \quad w_j = e^{\mu_j/\beta}.$$

First,

$$\max_j \xi_j$$

is Gumbel with scale β and location

$$\beta \log \left(\sum_j w_j \right) - \gamma \beta.$$

Second, the identity of the maximizing alternative follows the logit formula:

$$\Pr \left(\arg \max_j \xi_j = k \right) = \frac{w_k}{\sum_j w_j}.$$

After normalizing all weights by the common factor $e^{u^*/\beta}$, the variables Z_j , s_j , and W_j have weights x_j , λ_j , and $x_j c_j$, respectively.

The key identity underlying the closed-form expressions below is

$$x_j c_j = \min\{x_j, \lambda_j\}. \quad (\text{H.1})$$

H.2. Demand

H.2.1. Unassisted directed search

By the ordered-search result of [Choi et al. \(2018, Theorem 1\)](#), the optimal sequential-search procedure has an equivalent static representation. The consumer inspects products in decreasing order of $\varepsilon_j - p_j$ (or, equivalently, Z_j), stops when the highest surplus observed so far exceeds the highest remaining ranking score, and ultimately purchases

$$\arg \max_j W_j, \quad W_j = \min\{\bar{u}_j, u^*\} - p_j + \varepsilon_j. \quad (\text{H.2})$$

The reservation match value u^* solves

$$\int_{u^*}^{\infty} (\bar{u} - u^*) dF_{\bar{u}}(\bar{u}) = c_s. \quad (\text{H.3})$$

Because W_j is Gumbel with weight proportional to $x_j c_j$, the probability that product j is purchased is

$$X_j^0 = \frac{x_j c_j}{\sum_{s=1}^m x_s c_s} = \frac{\exp[(\min\{\bar{u}_j, u^*\} - p_j)/\beta]}{\sum_{s=1}^m \exp[(\min\{\bar{u}_s, u^*\} - p_s)/\beta]}. \quad (\text{H.4})$$

Under the baseline distribution $\bar{u} \sim U[-1, 1]$, equation (H.3) becomes

$$c_s = \frac{(1 - u^*)^2}{4},$$

and hence

$$u^* = 1 - 2\sqrt{c_s}.$$

H.2.2. Directed search with a recommendation

Suppose that product r is recommended and therefore inspected first. Naturally, once inspected, even if it is not purchased immediately, the recommended product remains available for later purchase. Thus, if the consumer continues searching, product r continues to compete with its full realized surplus s_r .

In the static representation of ordered search, forcing product r to be inspected first is equivalent to assigning it an infinite reservation value. Its capped surplus is therefore replaced by its full realized surplus, while all other products retain their original capped values:

$$\widetilde{W}_j^{(r)} = \begin{cases} s_r, & j = r, \\ W_j, & j \neq r. \end{cases}$$

The consumer purchases

$$\arg \max_j \widetilde{W}_j^{(r)}.$$

Define

$$\Theta_r = \lambda_r + \sum_{s \neq r} x_s c_s = \lambda_r + S - x_r c_r.$$

The conditional purchase probabilities are then

$$X_r^{(r)} = \frac{\lambda_r}{\Theta_r}, \quad X_j^{(r)} = \frac{x_j c_j}{\Theta_r}, \quad j \neq r. \quad (\text{H.5})$$

The consumer stops after inspecting the recommended product whenever

$$s_r > \max_{j \neq r} Z_j.$$

Accordingly, the probability that the recommendation fails this initial stopping test is

$$\begin{aligned} \Pr(F_r) &= \Pr\left(s_r \leq \max_{j \neq r} Z_j\right) \\ &= \frac{\sum_{j \neq r} x_j}{\lambda_r + \sum_{j \neq r} x_j} \\ &= \frac{\sum_{j \neq r} e^{(u^* - p_j)/\beta}}{e^{(\bar{u}_r - p_r)/\beta} + \sum_{j \neq r} e^{(u^* - p_j)/\beta}}. \end{aligned} \quad (\text{H.6})$$

If product r is recommended with probability ρ_r , as under the Boltzmann rule used in Section 11.1, the unconditional purchase probability is

$$X_j = \sum_{r=1}^m \rho_r X_j^{(r)}.$$

Restoring the consumer index, expected demand for product j is obtained by summing these probabilities across consumers.

Equations (H.4) and (H.5) provide a sharp characterization of when a recommendation changes demand. If

$$\bar{u}_r \leq u^*,$$

then

$$W_r = s_r \quad \text{and} \quad \lambda_r = x_r c_r.$$

Every term in (H.5) then coincides with its counterpart in (H.4). Recommending product r changes the order of inspection, but not the probability with which any product is ultimately purchased. If instead

$$\bar{u}_r > u^*,$$

then $W_r = Z_r$ under unassisted search, whereas forcing r to be inspected first replaces its capped value Z_r with its full realized surplus s_r . Its weight consequently rises from

$$x_r c_r = x_r$$

to

$$\lambda_r > x_r.$$

Thus, recommendations affect demand only when the recommended product has a systematic match value above the reservation level. This is the directed-search counterpart of the front-loading effect.

H.3. Expected number of inspections

To compute consumer surplus net of search costs, we also require the expected number of products inspected. The ordered-search representation yields simple closed-form expressions both without and with recommendations.

H.3.1. Unassisted directed search

Let T^0 denote the number of products inspected under unassisted directed search. Product j is inspected if and only if no other product stops the search before it is reached. In the static representation, this is equivalent to

$$Z_j \geq \max_{i \neq j} W_i.$$

By linearity of expectations and the Gumbel maximum property,

$$\mathbb{E}[T^0] = \sum_{j=1}^m \Pr \left(Z_j \geq \max_{i \neq j} W_i \right)$$

$$\begin{aligned}
&= \sum_{j=1}^m \frac{x_j}{x_j + \sum_{i \neq j} x_i c_i} \\
&= \sum_{j=1}^m \frac{x_j}{x_j + S - x_j c_j}.
\end{aligned} \tag{H.7}$$

Let

$$L = \{j : \bar{u}_j < u^*\}.$$

Any product with $\bar{u}_j \geq u^*$ stops the search as soon as it is inspected. The consumer can therefore inspect at most all products in L , followed by one product outside L . This gives the bound

$$T^0 \leq \min\{m, |L| + 1\}. \tag{H.8}$$

H.3.2. Directed search with a recommendation

Let $T^{(r)}$ denote the number of products inspected when r is recommended. The recommended product is inspected with probability one. A product $j \neq r$ is subsequently inspected if and only if

$$Z_j > \max \left\{ s_r, \max_{i \neq j, r} W_i \right\}.$$

This condition already implies that the recommended product has failed to stop the search, so no additional conditioning is required. The Gumbel maximum property then gives

$$\begin{aligned}
\mathbb{E}[T^{(r)}] &= 1 + \sum_{j \neq r} \Pr \left(Z_j > \max \left\{ s_r, \max_{i \neq j, r} W_i \right\} \right) \\
&= 1 + \sum_{j \neq r} \frac{x_j}{x_j + \lambda_r + \sum_{i \neq j, r} x_i c_i} \\
&= 1 + \sum_{j \neq r} \frac{x_j}{x_j + \lambda_r + S - x_j c_j - x_r c_r}.
\end{aligned} \tag{H.9}$$

When recommendations are probabilistic, the unconditional expected number of inspections is

$$\mathbb{E}[T] = \sum_{r=1}^m \rho_r \mathbb{E}[T^{(r)}].$$

Two implications of (H.9) are worth noting. First, suppose that

$$\bar{u}_r \leq u^*.$$

In this case, $\lambda_r = x_r c_r$, so the probability that any rival product is inspected is exactly the same as in the unassisted-search benchmark. Hence

$$\mathbb{E}[T^{(r)}] = \mathbb{E}[T^0] + 1 - \Pr(r \text{ is inspected under unassisted search}). \tag{H.10}$$

Recommending a product below the reservation level therefore increases expected search by precisely the probability that the consumer would not otherwise have inspected it, while leaving the inspection probabilities of all rival products unchanged. Second, if

$$\bar{u}_r > u^*,$$

then

$$\lambda_r > x_r c_r = x_r.$$

The recommendation not only guarantees that r is inspected, but also increases its weight in every subsequent stopping comparison. This reduces the probability that rival products are reached and shortens expected search relative to the purely mechanical effect described in (H.10).

H.4. Results

The effects of algorithmic recommendations on equilibrium prices under directed search are reported in Section 11. Here we report the corresponding changes in consumer surplus. Table H.1 shows that consumers lose when the market is small and roughly break even otherwise: the gains from better matching and shorter search are largely offset, and for $m = 7$ more than offset, by the price increases documented in the main text.

m	7	13	19
Collaborative filtering RS	−0.0153 (0.0016)	0.0762 (0.0010)	0.1244 (0.0008)
Content-based RS	−0.0153 (0.0016)	0.0769 (0.0010)	0.1253 (0.0008)

TABLE H.1: Changes in consumer surplus under directed search.

I. SELF-PREFERENCING

We vary the manipulation rate τ from 0 to 1 in increments of 0.1. For each value, the tables report the average profit of the firm supplying the favored product, $\bar{\pi}_f$, of competing firms, $\bar{\pi}_{-f}$, and of the industry as a whole, $\bar{\pi}$, together with the corresponding equilibrium prices, p_f , $\bar{p}_{-f}$, and $\bar{p}$, and consumer surplus. Results are reported under both conservative beliefs and rational expectations. The rows for $\tau = 0.2$ correspond to the results reported in Section 12 of the paper.

The favored firm’s profit is maximized at moderate manipulation rates, around 40% under conservative beliefs and 20% under rational expectations. By contrast, aggregate industry profits decline monotonically with τ . Under conservative beliefs, consumer surplus is roughly flat or slightly higher at low manipulation rates, as lower prices offset the deterioration in recommendation quality, but declines when manipulation becomes more extensive. Under rational expectations, by contrast, consumer surplus rises substantially because manipulation weakens the reliance effect.

τ	$\bar{\pi}_f$	$\bar{\pi}_{-f}$	$\bar{\pi}$	$\bar{p}_f$	$\bar{p}_{-f}$	$\bar{p}$	CS
0	0.0355	0.0355	0.0355	0.4501	0.4545	0.4541	0.2070
0.1	0.0543	0.0330	0.0347	0.4397	0.4427	0.4425	0.2096
0.2	0.0636	0.0309	0.0334	0.4111	0.4269	0.4257	0.2092
0.3	0.0671	0.0295	0.0324	0.3919	0.4145	0.4127	0.2036
0.4	0.0679	0.0286	0.0316	0.3805	0.4044	0.4025	0.1954
0.5	0.0675	0.0279	0.0309	0.3737	0.3958	0.3941	0.1860
0.6	0.0665	0.0274	0.0304	0.3690	0.3879	0.3865	0.1763
0.7	0.0655	0.0269	0.0298	0.3655	0.3807	0.3795	0.1663
0.8	0.0644	0.0264	0.0293	0.3625	0.3738	0.3729	0.1560
0.9	0.0633	0.0260	0.0288	0.3597	0.3670	0.3664	0.1453
1	0.0622	0.0255	0.0283	0.3570	0.3598	0.3596	0.1339

TABLE I.1: Profits, prices and consumer surplus under conservative beliefs.

τ	$\bar{\pi}_f$	$\bar{\pi}_{-f}$	$\bar{\pi}$	$\bar{p}_f$	$\bar{p}_{-f}$	$\bar{p}$	CS
0	0.0562	0.0559	0.0560	0.7155	0.7237	0.7231	-0.0012
0.1	0.0922	0.0488	0.0522	0.6652	0.6717	0.6712	0.0371
0.2	0.1032	0.0413	0.0460	0.5642	0.5917	0.5896	0.0925
0.3	0.1014	0.0363	0.0413	0.4952	0.5319	0.5291	0.1260
0.4	0.0950	0.0332	0.0379	0.4524	0.4878	0.4851	0.1434
0.5	0.0879	0.0311	0.0354	0.4251	0.4548	0.4525	0.1511
0.6	0.0813	0.0295	0.0335	0.4056	0.4286	0.4268	0.1534
0.7	0.0755	0.0282	0.0319	0.3904	0.4071	0.4059	0.1520
0.8	0.0706	0.0272	0.0305	0.3778	0.3893	0.3884	0.1480
0.9	0.0662	0.0263	0.0294	0.3669	0.3739	0.3733	0.1419
1	0.0622	0.0255	0.0283	0.3570	0.3598	0.3596	0.1339

TABLE I.2: Profits, prices and consumer surplus under rational expectations.